DRUMPANEL · PROJECT PLAN

DrumPanel project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First project plan (kit 1.8 and 1.9.1 upgrade of a TRL 3 repo); stage P3 reached; workstreams, gates, decision requests and references

DrumPanel project plan

DrumPanel reached TRL 3 on 2026-10-03 and stage P3 (design outputs) on 2026-10-24 with the kit 1.9.1 upgrade: pipeline documents, two concept options both modelled to D3, and the chosen design's model detailed part by part. The next gate, P4 (verification, TRL 4), is on hold while the portfolio cap is P3. Two decisions are requested from Amish: the concept carried forward (option A recommended) and the flash point limit for accepting drums (keep 60 °C recommended). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
STEM check	Design lead	DMP-STM-001	Pass (2026-10-24); independent review recorded; five fixes made under the standing instruction (F1 to F5)
Concept options	Amish Chadha	DMP-OPT-001, DMP-DDR-005	Option A recommended; proposed, awaiting Amish
Requirements	Design lead	DMP-URS-001, DMP-PRS-001, DMP-RTM-001	Done on paper; partner review to do
Calculations	Design lead	DMP-CAL-001	Done (v0.3)
Detailed CAD	Design lead	`cad/src/model.py`, both options at D3, STEP, drawings	Done (`cad/fidelity.json`)
Build plan and BOM	Design lead	DMP-BLD-001, `bom/bom.csv`	Done (plan, not yet built)
Regulatory, risk, market	Design lead	DMP-REG-001, DMP-RSK-001, DMP-MKT-001	First pass done; country rules to assess with a partner
Photoreal renders and options picture	Portfolio coordinator	`media/options.png` , `media/render-*.png` from the D3 models	Queued (rendered on Amish's Mac)
Co-design partner	Amish Chadha	A cooperative of drum artists and a vocational welding school	Not started (candidates in DMP-PRB-001)
Verification (TRL 4)	Design lead with a partner	Test plan and trials: vapour check procedure, slip roll trial pass, scrap-chime punch test, timed half-day trial	On hold (stage cap P3)

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead (Claude with Amish's review)	P0 to P3 documents, model, drawings, build plan	Done	0
Prototype parts (TRL 4, on hold)	One bench set	A town fabricator, about two weeks (estimate)	3,566 (BOM)
Gas detector training and calibration gas (TRL 4)	A person trained for each session	One day with the supplier (estimate)	Included in BOM line 4

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Done 2026-10-24 (the design itself dates from 2026-10-03)
P1 Definition	URS, PRS draft, RSK, MKT, REG complete	Done 2026-10-24
P2 Design inputs	PRS targets, CAL, RTM, DDR, STEM check closed, FEA closed	Done 2026-10-24
P3 Design outputs	Model at D3 for both options, STEP, DWG, priced BOM, BLD, DEC, design FMEA, RTM	Done 2026-10-24
P4 to P6	Tests, field trials, launch	On hold

Next steps and decisions requested

- Amish: decide the concept carried forward (DMP-DEC-001, open decision 1; option A recommended) and the drum acceptance flash point (open decision 2; keep 60 °C with the vapour check recommended).
- Portfolio coordinator: render `media/options.png` and the photoreal views from the D3 models, then the storefront cards.
- When the portfolio cap allows TRL 4: approach the first co-design candidate, build to DMP-BLD-001 and run the first checks, starting with the vapour check procedure and the slip roll trial pass.

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